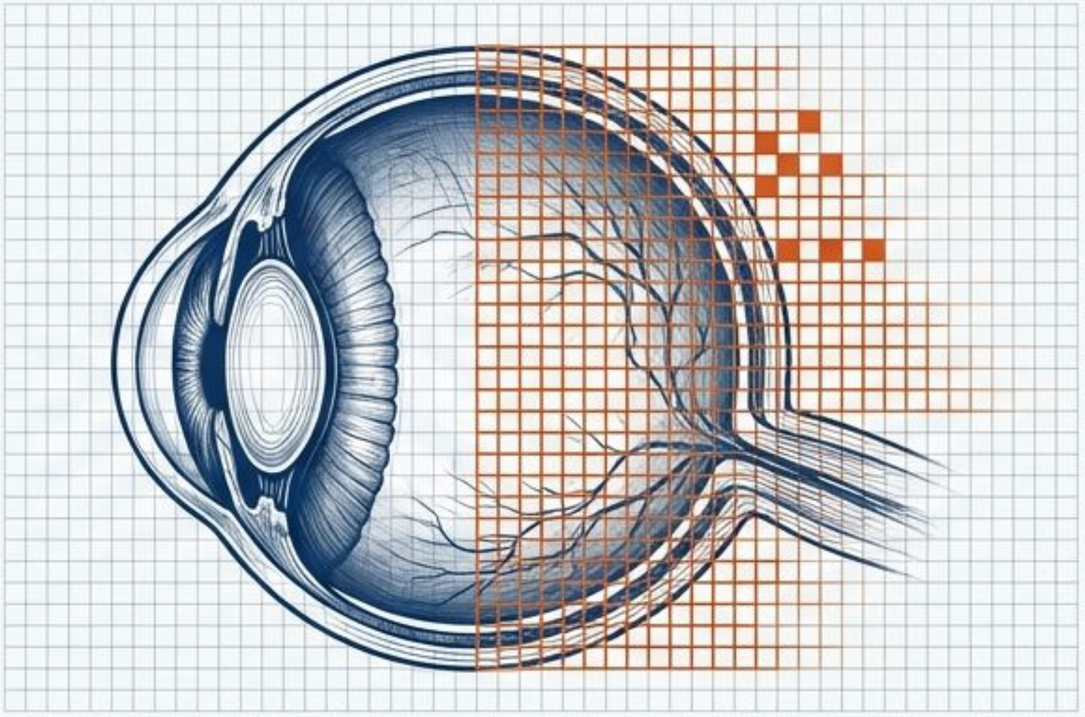
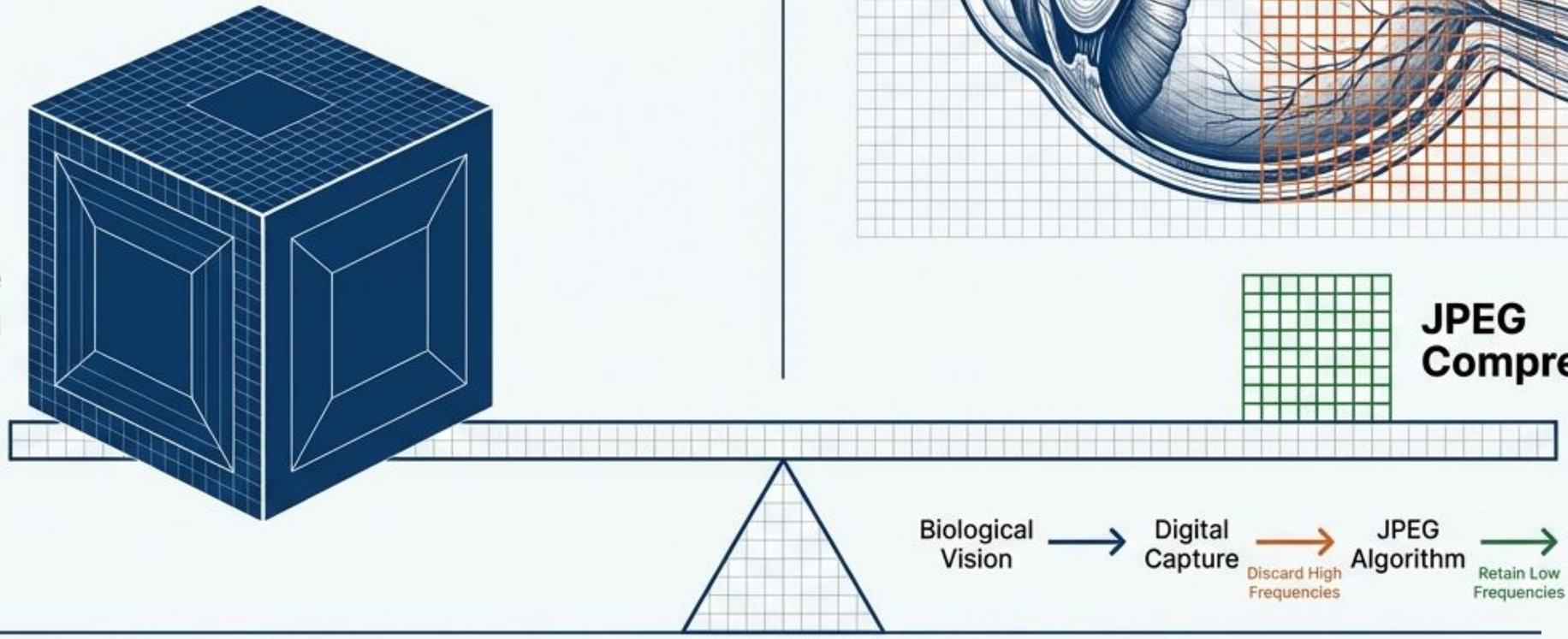
Hacking Human Vision for Digital Efficiency

- Industry standard for digital media compression
- **Goal:** Drastically reduce size, maintain structure
- **Method:** Inherently lossy compression architecture
- **The Secret:** Exploits biological limits of the eye
- **Execution:** Eliminates unnoticeable high-frequency detail

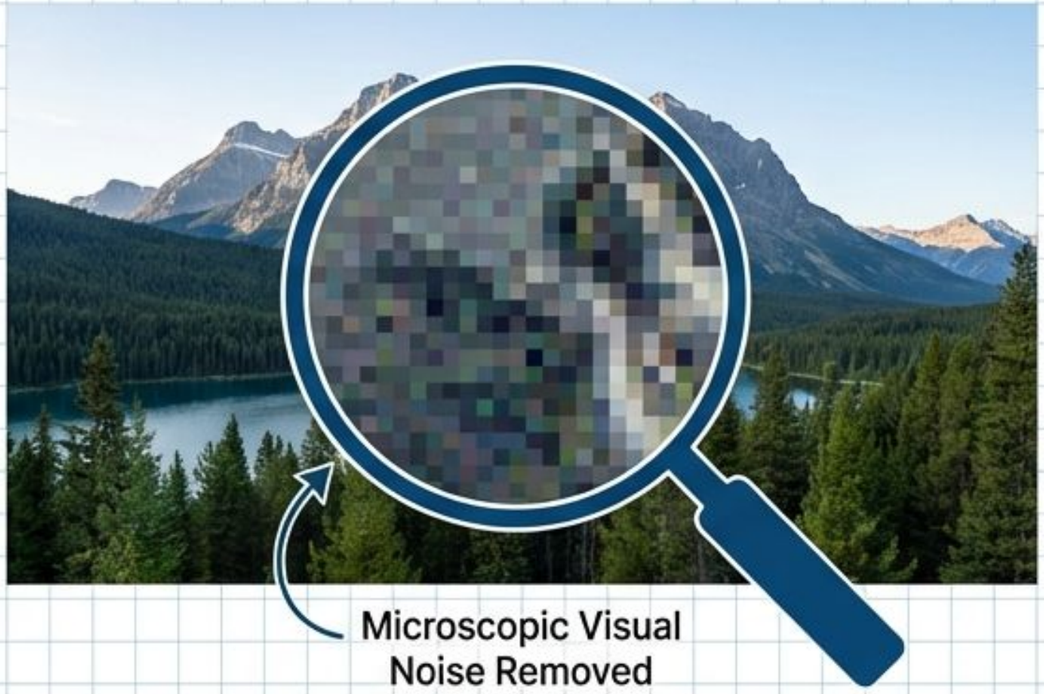




Biological Vision → Digital Capture → (Discard High Frequencies) JPEG Algorithm → (Retain Low Frequencies) Efficient Storage/ Transmission

The Seven-Step Digital Refinery

Raw Structural Integrity



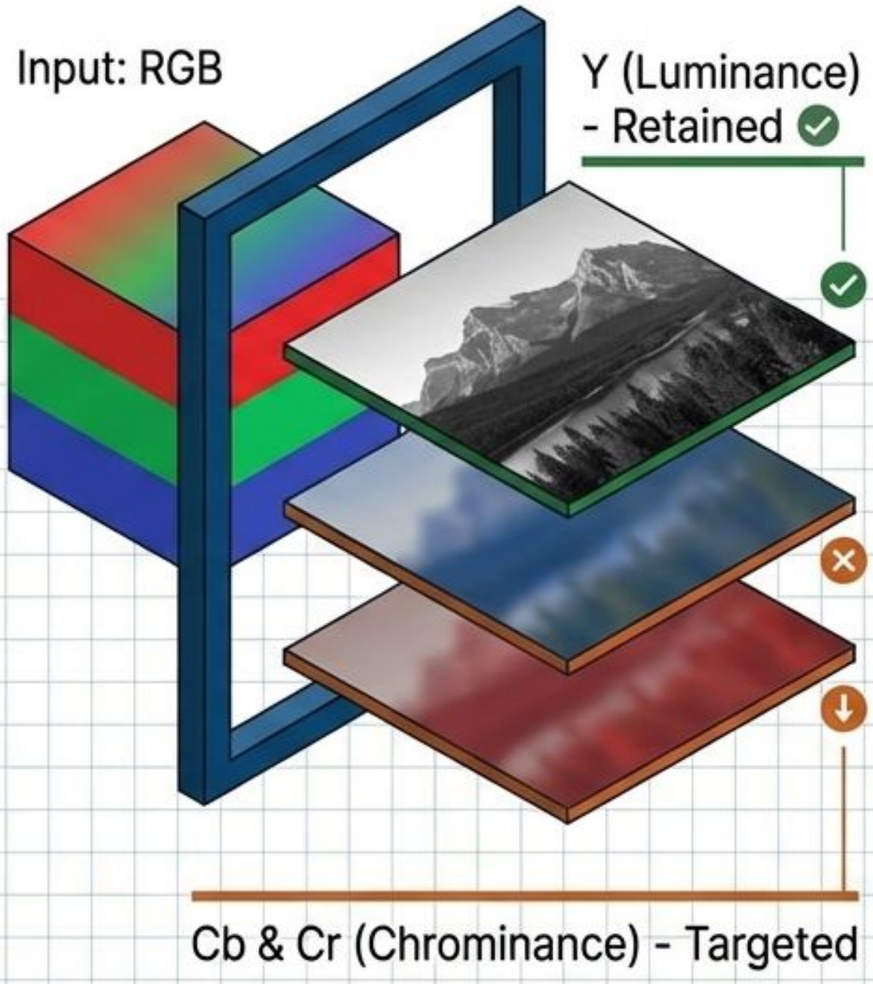
Microscopic Visual Noise Removed

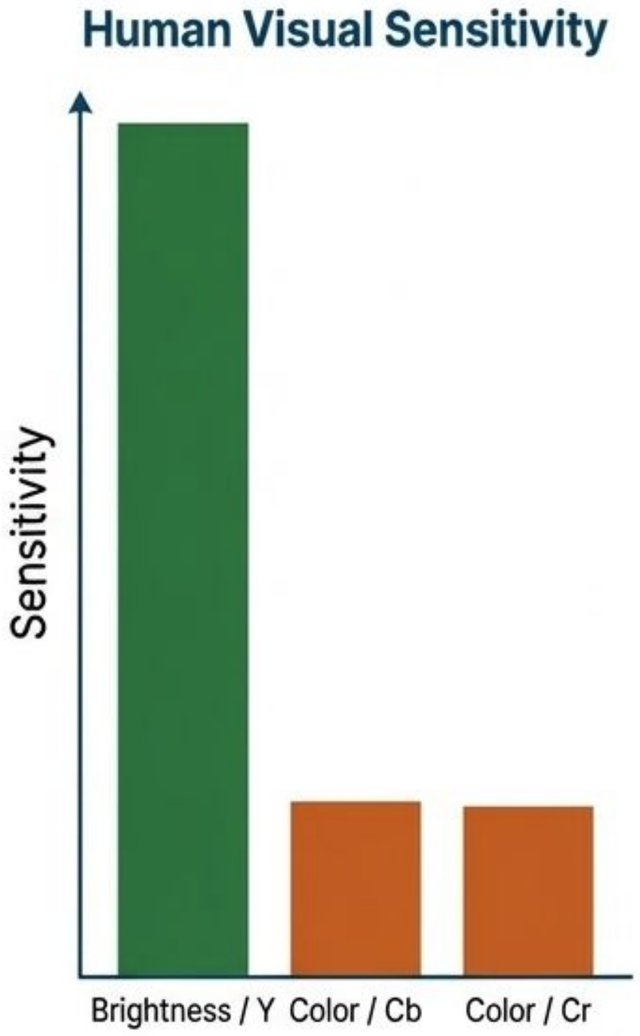
Preserves core brightness and smooth areas while executing a strict mathematical pipeline to discard imperceptible noise.

1. Color Space Conversion → 2. Chrominance Downsampling → 3. 8×8 Block Splitting → 4. Discrete Cosine Transform → 5. Quantization (Data Loss) → 6. Zig-Zag Scanning → 7. Huffman + RLE Encoding

Step 1: Translating Hardware RGB to Perceptual YCbCr

- **Native State:** Sensors capture Red, Green, Blue
- **Transformation:** Algorithm translates RGB to YCbCr
- **Y Channel:** Luminance (Core Brightness)
- **Cb/Cr Channels:** Chrominance (Color Information)
- **Biological Hack:** Human vision is acutely sensitive to brightness shifts, but blind to subtle color loss.

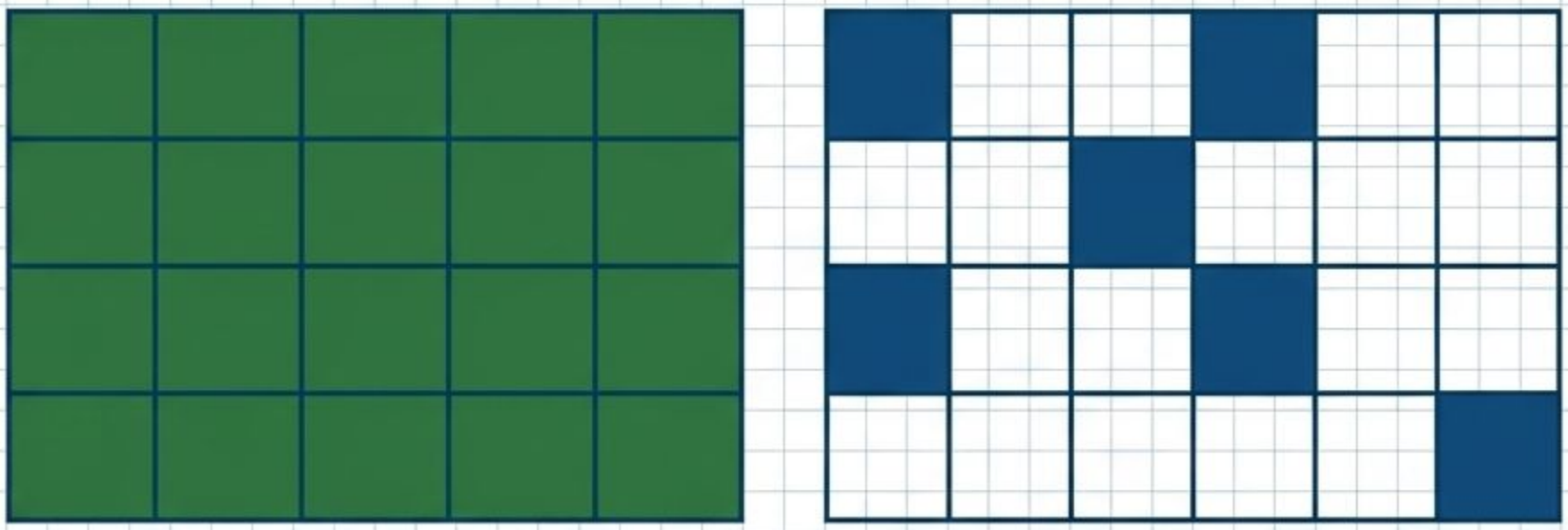




Step 2: Reducing Color Data via Downsampling

- **Targets only** the isolated Cb and Cr color channels
- **Physically halves** spatial color resolution
- **Luminance** remains entirely untouched
- **Massive** initial data savings with zero perceived loss

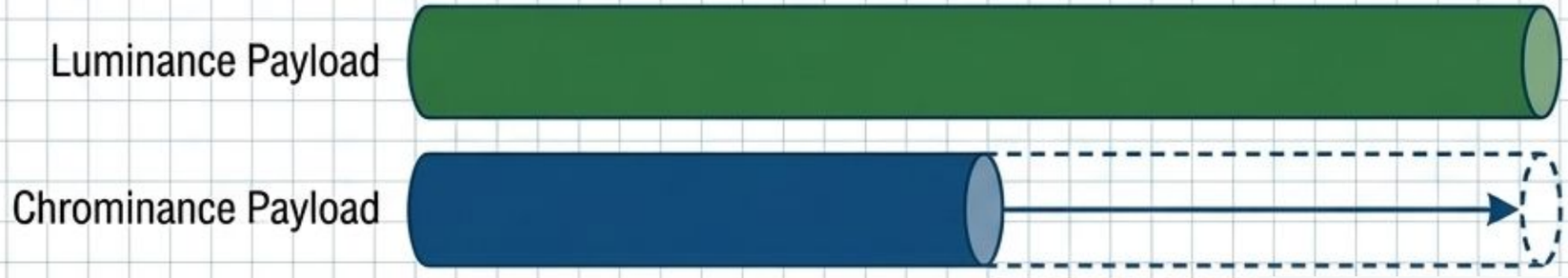
The Grid Comparison



Luminance (Y) - Native Resolution
100% of spatial resolution retained.

Chrominance (Cb/Cr) - 4:2:0 Downsampling
Color resolution physically quartered.

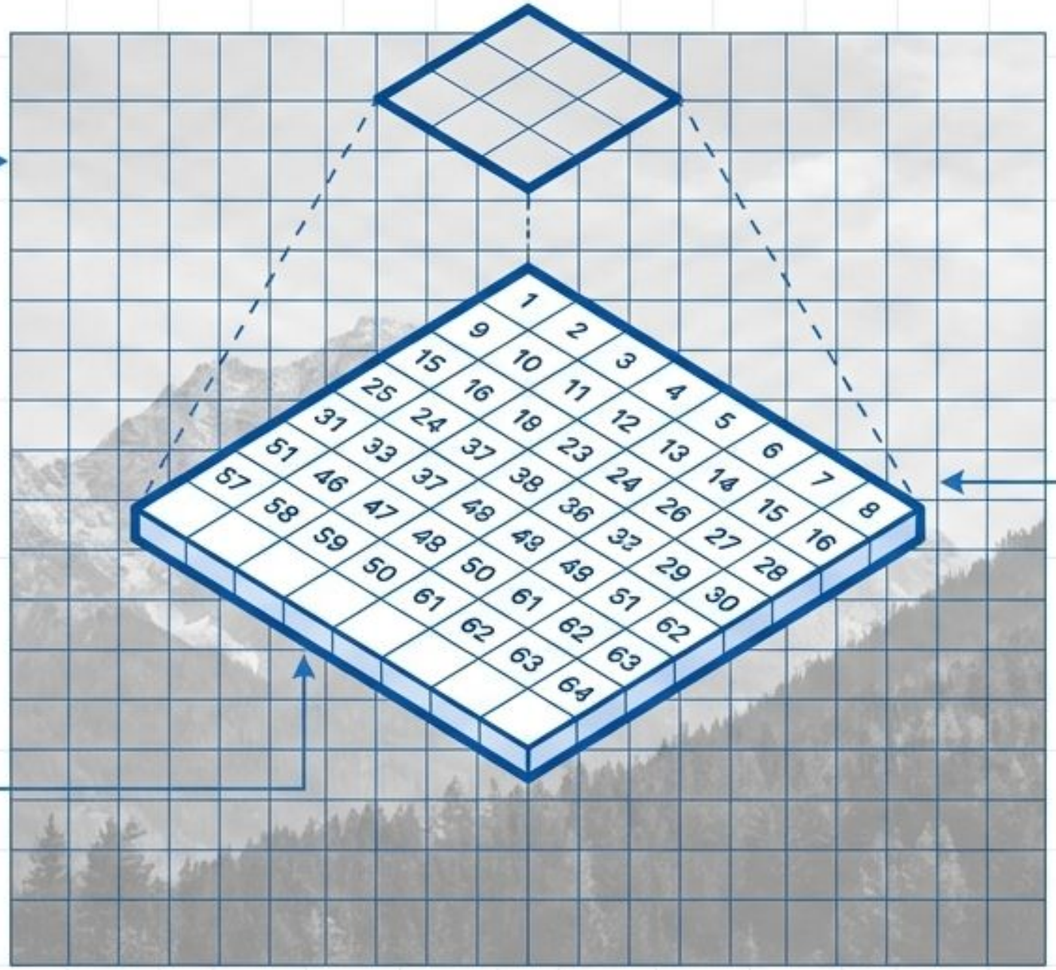
Data Volume Cylinders



Step 3: Isolating the Image into 8×8 Macro Blocks

Grid Subdivision

The entire image is strictly segmented into a rigid mathematical grid.



Independent Processing

The algorithm processes every 64-pixel block entirely on its own, creating the hallmark 'blocky' artifacts at extreme compression levels.

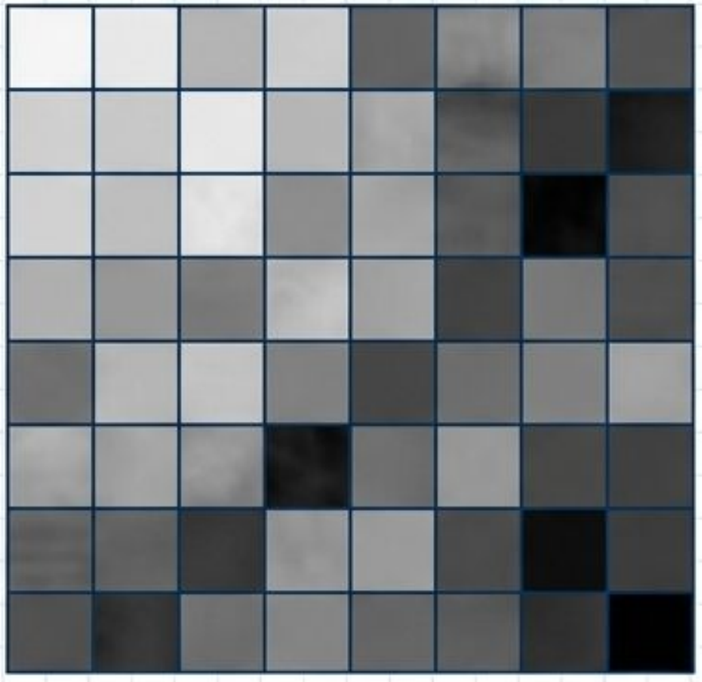
Standard Operational Unit

Each unit contains exactly 8x8 pixels (64 pixels total). This is the foundation of JPEG math.

Step 4: Applying the Discrete Cosine Transform (DCT)

- Translates spatial pixels into frequency waves
- Separates smooth areas from sharp details
- Prepares the matrix for data truncation

Spatial Domain (Raw Pixels)



Discrete Cosine Transform

Frequency Domain (Coefficients)

DC Coefficient: Macro Brightness (Retained)

AC Coefficients: Micro Textures & Edges (Expendable)

904.5	13.2	-45.1	1.8	4.8	13.6	-15.7	1.8
133.3	-21.6	-22.6	-14.8	-3.7	33.2	3.4	12.9
101.8	-33.5	-43.5	-12.0	-3.2	44.8	26.6	14.8
94.8	23.8	16.8	-18.5	-11.5	-2.5	-16.9	2.1
47.4	-13.6	-30.3	-19.6	38.3	57.5	23.1	1.7
-16.7	-20.5	-28.9	-1.8	28.4	46.2	12.5	8.6
				33.0	34.3	33.9	2.3
-45.3	-16.2	-45.2	3.5	28.6	16.1	12.9	1.8

Low Frequency (Crucial Data)

High Frequency (Targeted for Deletion)

Step 5: Irreversible Data Truncation (Quantization)

DCT Matrix

904.5	13.2	-45.1	1.8	4.8	13.6	-15.7	1.8
133.3	-21.6	-22.6	-14.8	-3.7	33.2	3.4	12.9
101.8	-33.5	-43.5	-12.0	-3.2	44.8	26.6	14.8
94.8	23.8	16.8	-18.5	-11.5	-2.5	-16.9	2.1
47.4	-13.6	-30.3	-19.6	38.3	57.5	23.1	1.7
-16.7	-20.5	-28.9	-1.8	28.4	46.2	12.5	8.6
33.0	34.3	33.9	2.3	28.6	16.1	12.9	1.8
-45.3	-16.2	-45.2	3.5	28.6	16.1	12.9	1.8

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Quantization Table

16	11	10	16	24	40	51	61
12	12	14	19	26	58	60	55
14	13	16	24	40	57	69	56
14	17	22	29	51	87	80	62
18	22	37	56	68	109	103	77
24	35	55	64	81	104	113	92
49	64	78	87	103	121	120	101
72	92	95	98	112	100	103	99

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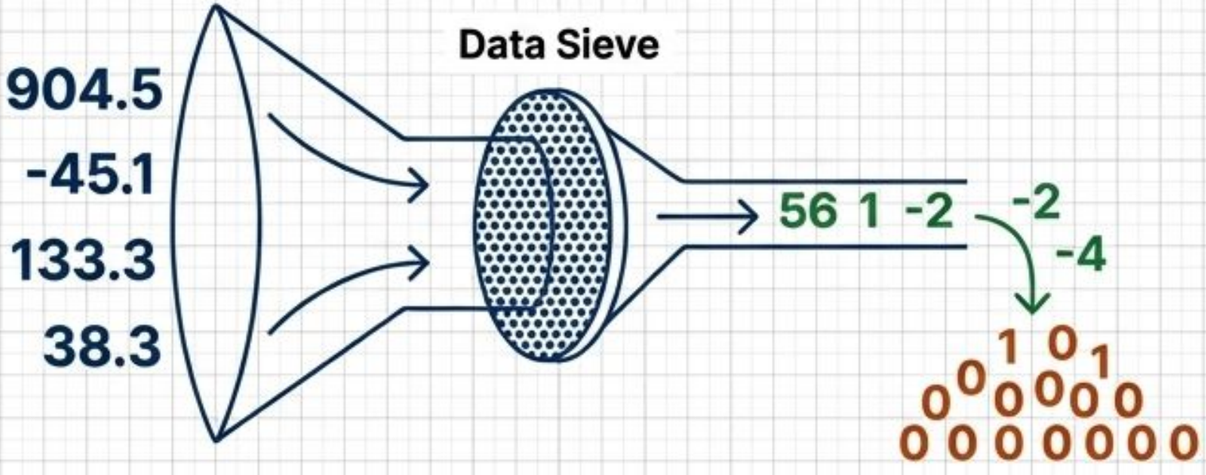
Resulting Matrix

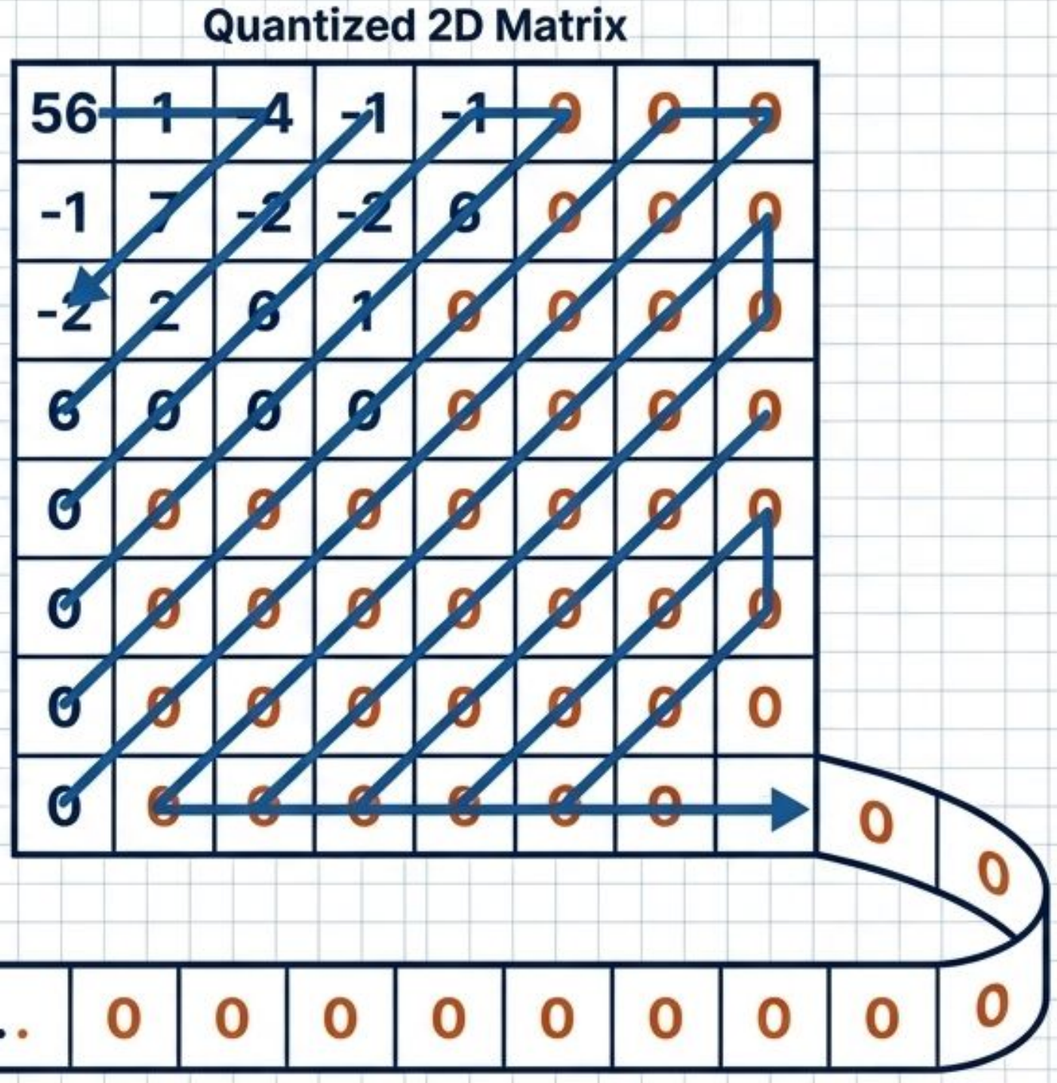
56	1	-4	0	0	0	0	0
11	-1	-1	0	0	0	0	0
7	-2	-2	0	0	0	0	0
6	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

- The most critical step for file size reduction
- Divides the DCT matrix by a standardized table
- Mathematically forces high-frequency values to zero
- IRREVERSIBLE: This is the exact point of data destruction



Step 6: Aligning Data via Zig-Zag Scanning

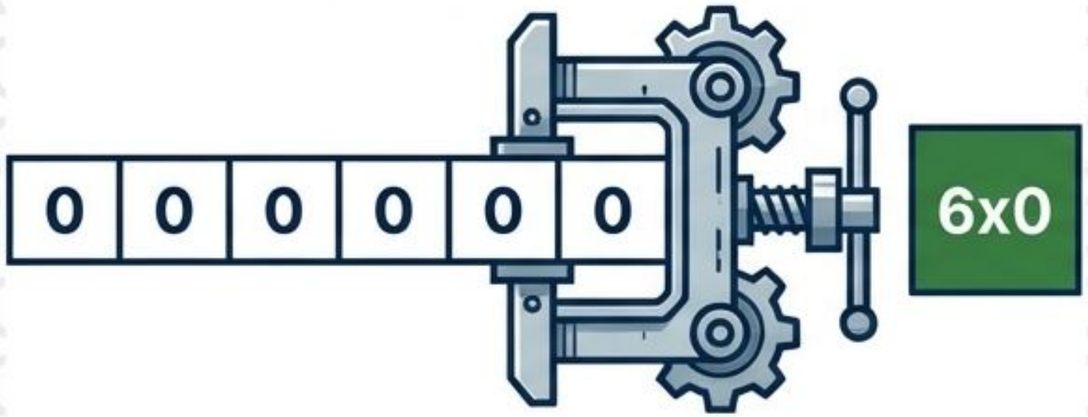
- Unspools the 2D matrix into a 1D linear sequence
- Scanning initiates at the crucial top-left DC value
- Sweeps diagonally toward the bottom-right
- Purpose: Groups all newly created zeros together into a continuous block for optimal compression.



56	1	-4	-1	-1	7	-2	-2	6	1	0,	…	0	0	0	0	0	0	0	0	0

Step 7: Final Binary Packaging (RLE & Huffman)

Phase A: Zero Compression (RLE)



Collapses continuous strings of grouped zeros into hyper-efficient coordinate markers.

Phase B: Frequency Optimization (Huffman)

Highly Frequent Value	→ 10	2-bit
Extremely Rare Value	→ 00110101	8-bit

Assigns the absolute shortest binary codes to the most common values, drastically reducing the final file weight.

Final Output: A highly compressed, mathematically refined .JPEG binary payload.

Academic Cheat Sheet: Key Exam Takeaways

Pipeline Phase	Key Technology / Action	Data Impact
Overall Architecture	Hacks human vision to discard noise	Inherently Lossy
Core Geometry	Segments image into 64-pixel units	8x8 Pixel Blocks
Mathematical Engine	Translates pixels to frequency waves	Discrete Cosine Transform (DCT)
Point of Data Destruction	Forces high frequencies to zero	⚠ Quantization (Irreversible Loss)
Final Packaging	Compresses zeros and optimizes bits	✅ Huffman Coding + RLE (Lossless)

Summary: JPEG relies heavily on Quantization to permanently eliminate unnoticeable high frequencies, dramatically reducing file size at the cost of microscopic structural integrity.